

Greg Chism

From social insects to machine learning: data science grounded in messy, real-world systems.

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Beyond the desk



National parks. The long walk-in, the early start.



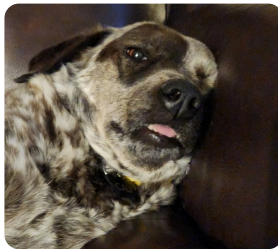
Fly fishing. Mostly an excuse to stand in cold water and think.

Curiosity about how small agents build big structures is the through-line. It started with animals and never really stopped.

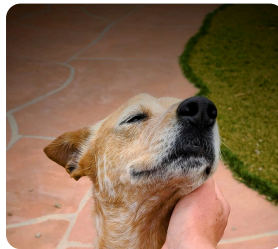
The crew. Three dogs with three incompatible opinions about the office chair.



Finn



Cali



Miles

How I got here



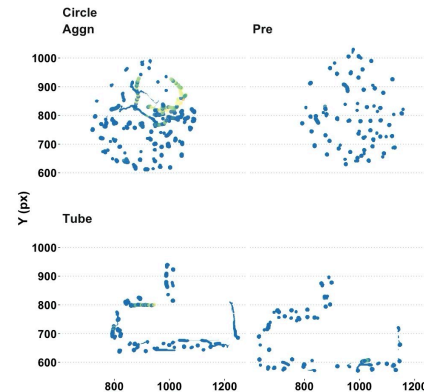
Social spiders

UCSB · Undergraduate



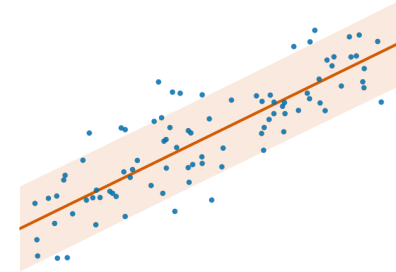
Social insects

Arizona · Ph.D.



Nest analytics

Where the data turn began



Data science

Arizona · College of
Information Science

The turn: measuring *how ant colonies use the space they build* meant tracking hundreds of individuals for days at a time. The bottleneck stopped being the microscope and became the analysis, so I went to fix the analysis and never came back.

Two research programs

01 · Healthcare analytics: measurement you can trust

02 · AI-assisted research workflows: pipelines that scale

Healthcare: the question

Does chronic pain distort depression screening?

The PHQ-8 counts somatic symptoms: **sleep, fatigue, appetite, psychomotor slowing**. Those are also everyday features of chronic pain, which raises the worry of *criterion contamination*. The tool might be counting pain and recording it as depression.

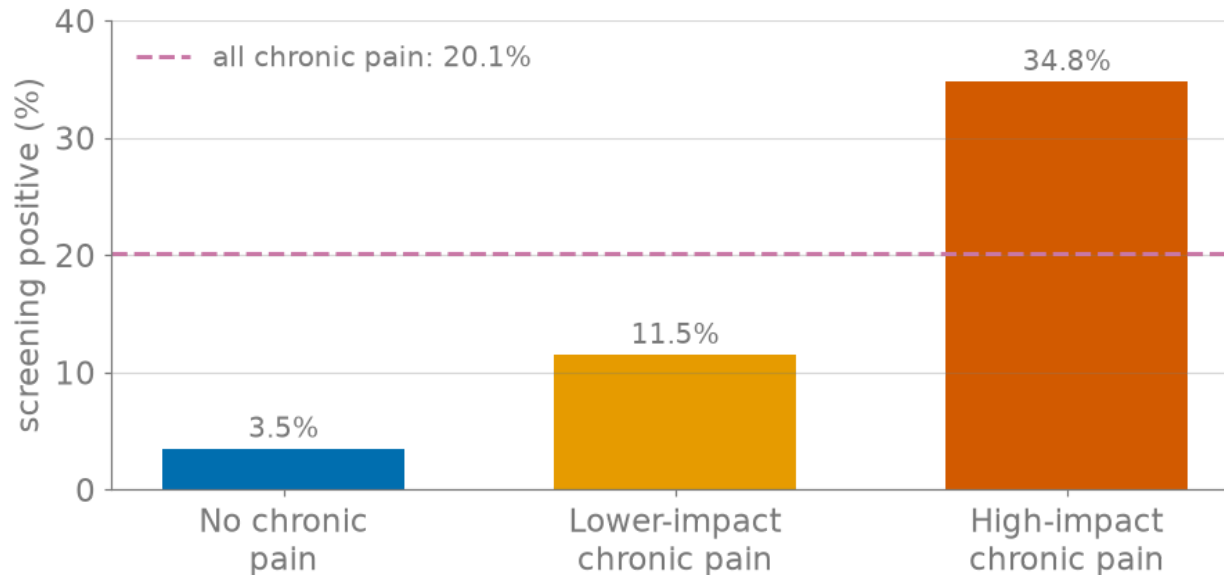


Figure 1: Positive PHQ-8 screens (≥ 10) by pain impact, 2019 NHIS (n = 30,983). Dashed line: all chronic pain combined (20.1%), which is the union of the two CP bars rather than a fourth group.

The finding: it doesn't. The gap is real rather than an artifact of the instrument.

Healthcare: why it's trustworthy

Check	No CP	CP
Cronbach's α	0.82	0.85
McDonald's ω	0.86	0.89
CFI	0.994	0.993
TLI	0.991	0.991

Invariance holds at the configural, metric, and scalar levels. **No differential item functioning.**

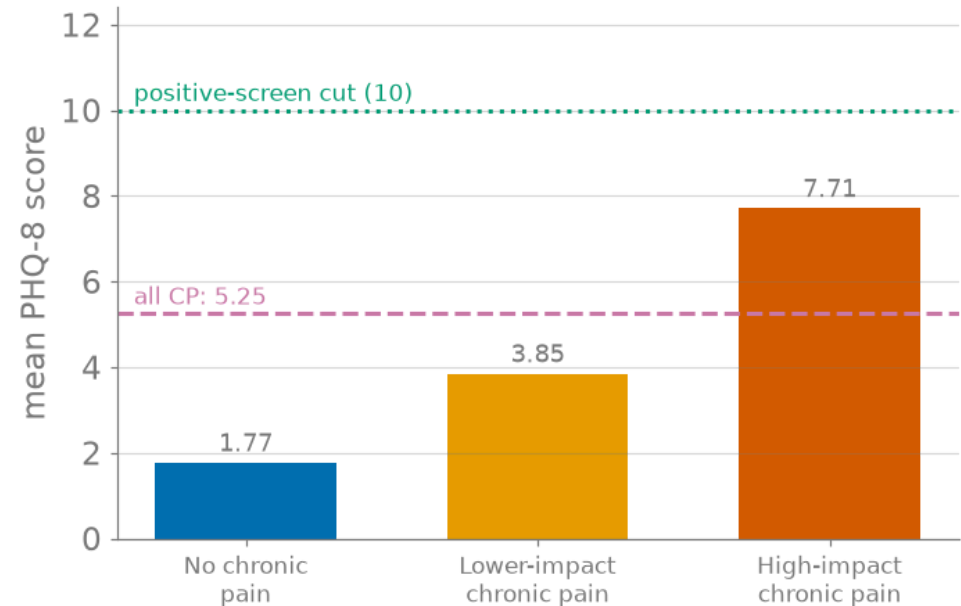


Figure 2: Mean PHQ-8 by pain impact. Even the high-impact mean (7.71) sits below the ≥ 10 screening cut point.

Bottom line: the PHQ-8 measures the *same thing* in both groups, so cross-group comparison is valid, and the 5-to-10 $\times$ disparity is a real clinical difference rather than a measurement artifact.

AI workflows: the system

3,681 state bills (OpenStates + LegiScan, 2017–) → **2,364 AI-relevant** (64%), 51 jurisdictions, **14-category** governance taxonomy, mean 2.86 categories per bill.

Six-model ensemble, zero-shot. Three Claude models (Sonnet, Opus, Haiku) and three open-source (Qwen-2.5-72B, Llama-3.1-70B/8B). F1-weighted majority vote, with confidence tiers from raw six-model agreement.

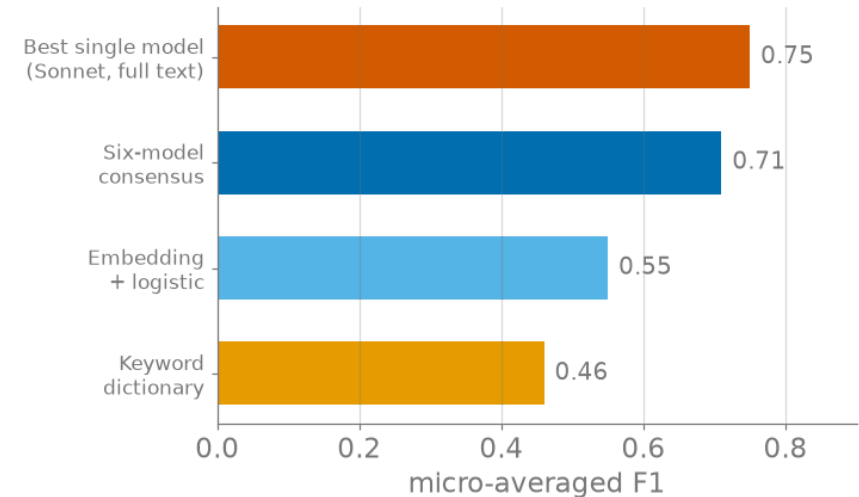


Figure 3: F1 on the $n = 20$ human gold set. The ensemble does *not* beat Sonnet alone. What it buys is calibrated confidence.

Honest caveat: the gold set is **$n = 20$, single-coder**. That fixes the *ordering* of methods, but not tight per-category intervals.

AI workflows: what the data shows

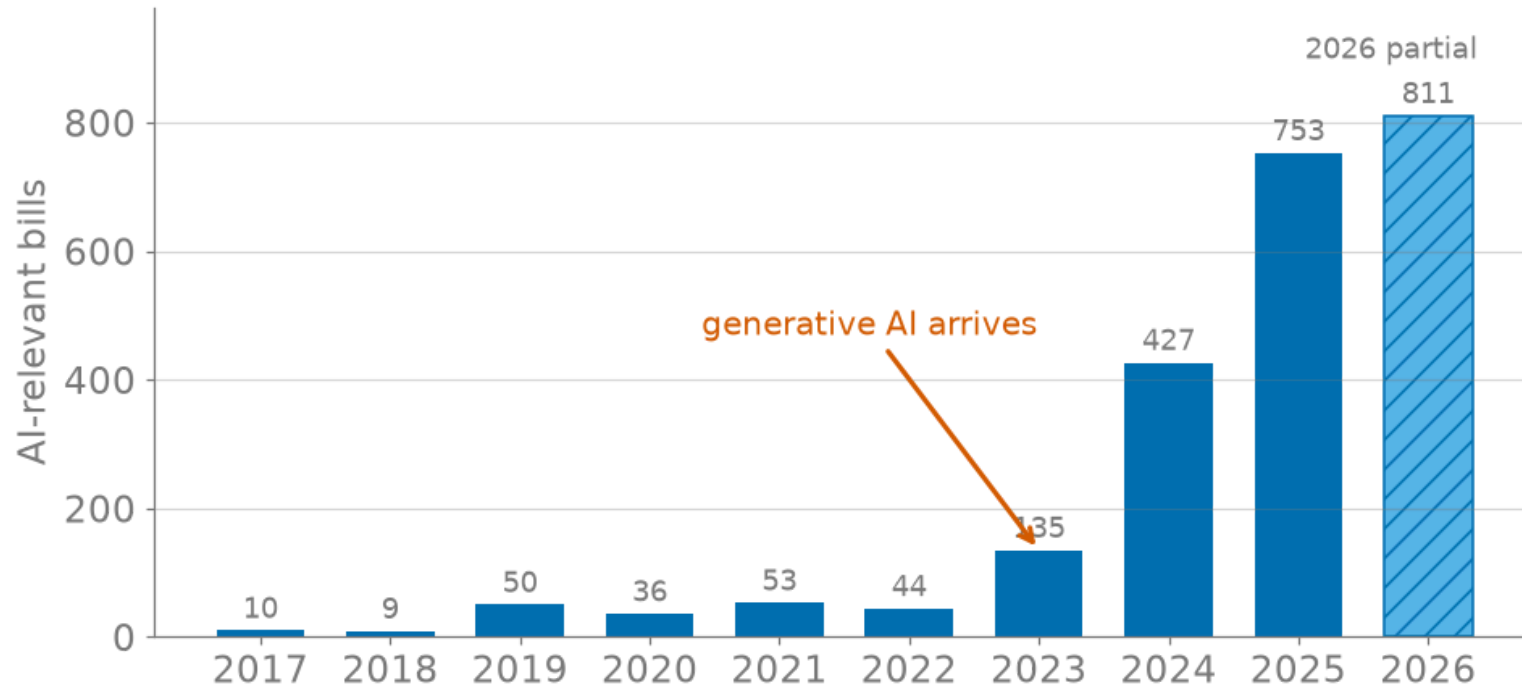


Figure 4: AI-relevant bills by year of first action. 2026 (hatched) is a partial year, with the corpus frozen 2026-07-10, and it has already passed all of 2025.

Enactment: 17.7% of 1,935 decided bills. Narrow bills pass: Education 22.9%, Criminal Use 22.8%. Accountability stalls: Private Right of Action 12.8%, Industry Use 12.8%, Taxes 5.7%.

Cox model (2,250 bills, 342 events): sponsor count **HR 1.23**; Disclosure **HR 0.80** and Private Right of Action **HR 0.79** *slow* enactment. The mechanisms with teeth are the ones that stall.

INFO 521

**Probabilistic and from-scratch · healthcare-grounded ·
reproducible by design**

Learning outcomes

By the end of this course, you will be able to:

1. **Identify and evaluate** machine-learning approaches appropriate for regression, classification, clustering, and dimensionality-reduction problems.
2. **Implement** machine-learning algorithms in Python using industry-standard tools and workflows.
3. **Compare and interpret** model performance using appropriate evaluation metrics, validation techniques, and statistical reasoning.
4. **Apply** machine-learning methods to real-world datasets to develop predictive and descriptive models.
5. **Communicate** results, limitations, and recommendations through written, visual, and technical presentations.

The seven-module arc

loss → *likelihood* → *posterior* → *approximation* → *discovery*

	Module	The idea
M1	Linear Least Squares	the model, the normal equations
M2	Generalization	CV, bias–variance, ridge
M3	Likelihood & MLE	least squares, re-earned
M4	Bayesian Inference	priors, posteriors, conjugacy
M5	Approximate Inference	logistic + Newton, Laplace, MCMC
M6	SVM & Clustering	max-margin, k-means, GMM + EM
M7	PCA	eigen-decomposition of the covariance

One object runs through all of it: $X^T X$. The least-squares bowl (M1), the likelihood's curvature and Fisher information (M3), the Laplace posterior covariance (M5), the covariance you eigen-decompose (M7).

How the course works

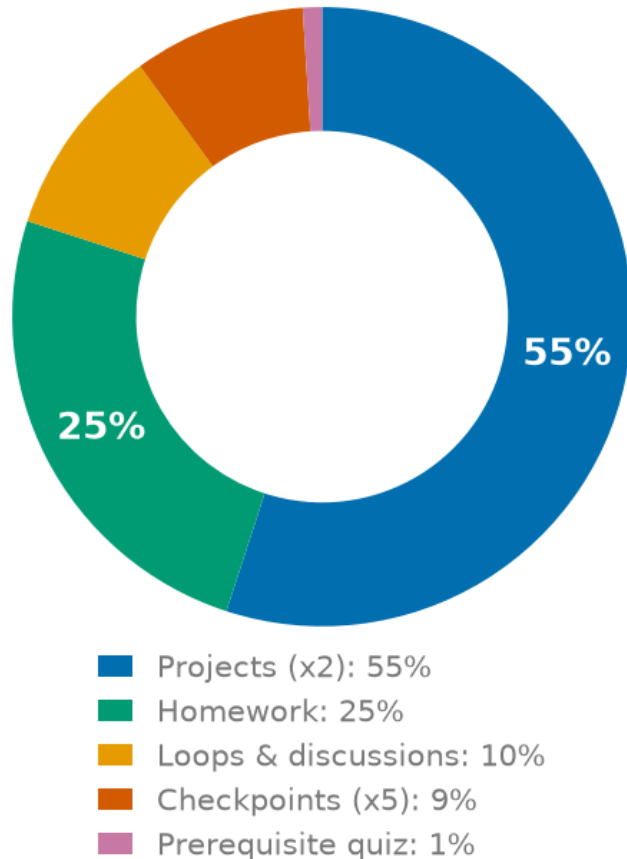


Figure 5: Grade weighting. Projects and homework together make up 80%, the work that looks most like real practice.

- **Specs grading.** Work is **Pass** or **Not-Yet**, never partial credit. Autograded correctness resubmits **without limit**; human-reviewed components draw on a small pool of revision tokens.
- **Two projects, one seam.** *One Problem, Three Lenses* → *From Inference to Discovery*. Project 1 closes on the **conjugate Gaussian posterior**; Project 2 opens by **binarizing the outcome to break conjugacy**, which is exactly what forces approximate inference.
- **Interactive weekly activities.** Each module ships a browser tool you use in lecture and again in the peer loop.
- **Peer-engagement loops, four stages:** tool + mastery check → reflection → review **every** teammate → respond and revise. Credit needs all four.

Where this lands

The clinical data you will model in INFO 521 is **the same kind of data I study**: national health survey data, self-reported instruments, real measurement error, and consequences attached to the answer.

So the questions this course trains you to ask are not academic:

- Is the gap I measured **real**, or an artifact of the instrument?
- Does my model do better than the **simplest thing that could work**?
- What does my method get wrong, and **who does that cost**?

Every method in this course is a way of being honest about uncertainty.

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